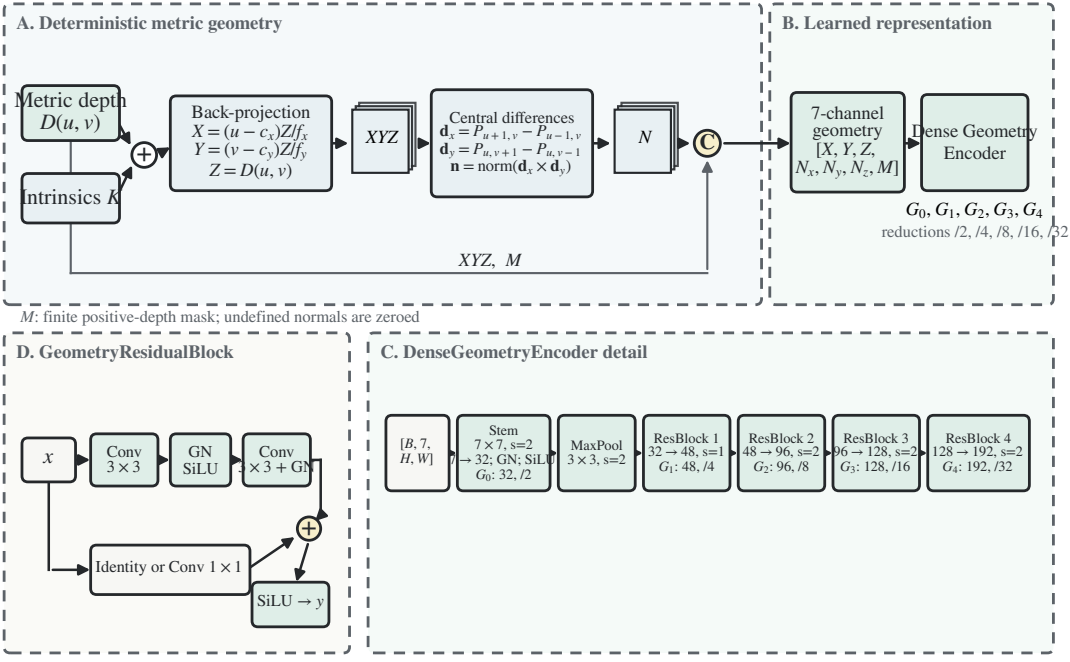
A. Deterministic metric geometry
Metric depth
$D(u, v)$
Intrinsics K
Back-projection
$X = (u - c_x)Z/f_x$
$Y = (v - c_y)Z/f_y$
$Z = D(u, v)$
XYZ
Central differences
$\mathbf{d}_x = P_{u+1,v} - P_{u-1,v}$
$\mathbf{d}_y = P_{u,v+1} - P_{u,v-1}$
$\mathbf{n} = \mathrm{norm}(\mathbf{d}_x \times \mathbf{d}_y)$
N
C
XYZ, M
B. Learned representation
7-channel geometry
$[X, Y, Z, N_x, N_y, N_z, M]$
Dense Geometry Encoder
G_0, G_1, G_2, G_3, G_4
reductions /2, /4, /8, /16, /32
M: finite positive-depth mask; undefined normals are zeroed
D. GeometryResidualBlock
x
Conv 3 × 3
GN SiLU
Conv 3 × 3 + GN
Identity or Conv 1 × 1
SiLU → y
C. DenseGeometryEncoder detail
[B, 7, H, W]
Stem
7 × 7, s=2
→ 32; GN; SiLU
G_0: 32, /2
MaxPool
3 × 3, s=2
ResBlock 1
32 → 48, s=1
G_1: 48, /4
ResBlock 2
48 → 96, s=2
G_2: 96, /8
ResBlock 3
96 → 128, s=2
G_3: 128, /16
ResBlock 4
128 → 192, s=2
G_4: 192, /32